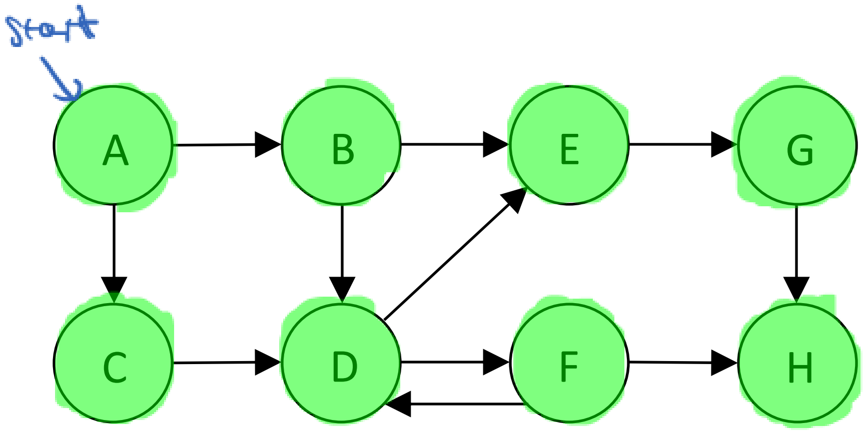
In what order will we visit the nodes in the following graph during a breadth-first traversal with starting node A?



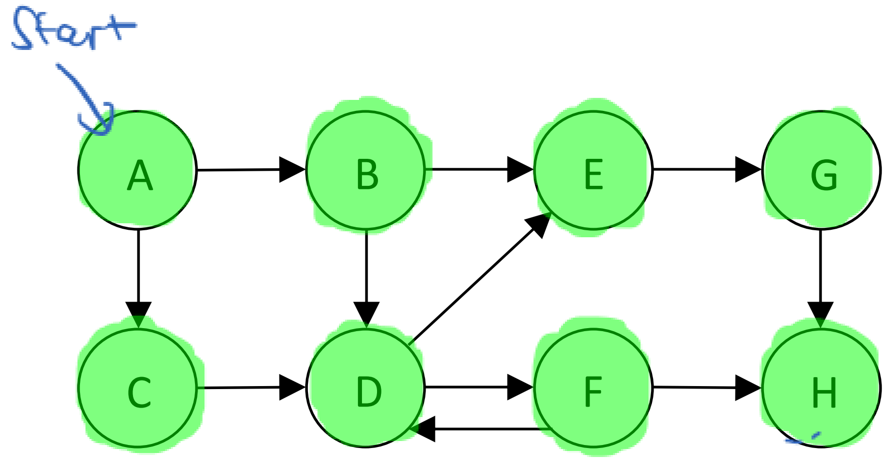
queue :

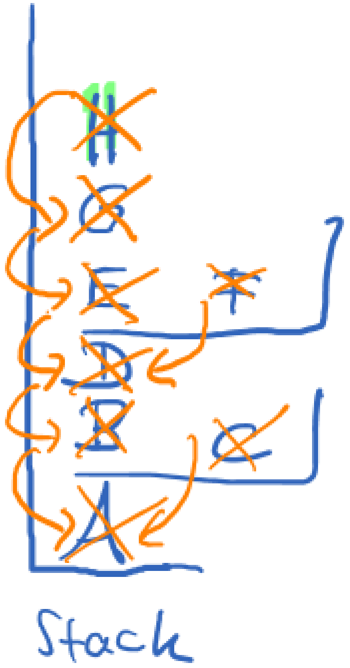
← out ← in

~~A~~, ~~B~~, ~~C~~, ~~D~~, ~~E~~, ~~F~~, ~~G~~, ~~H~~

visiting sequence: A, B, C, D, E, F, G, H

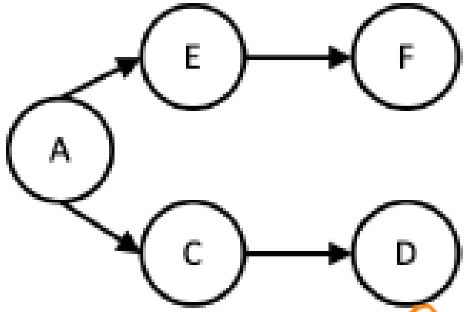
In what order will we visit the nodes in the following graph during a depth-first traversal with starting node A?





visiting sequence: A, B, D, E, G, H, F, C

Which of the following sequences of nodes is a valid breadth-first traversal of this graph?

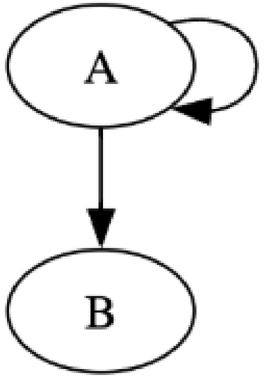


A. ~~A, E, F, C, D~~
B. ~~A, F, E, C, D~~
C. ~~D, C, A, E, F~~
D. A, E, C, F, D
E. ~~A, D, C, E, F~~

A, C, E, D, F

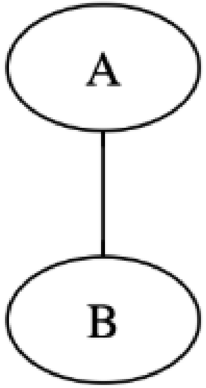
Which of the following is an example of an acyclic graph?

~~a.~~



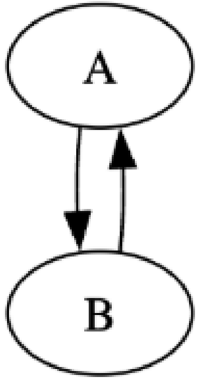
Cycle: A A

~~b.~~



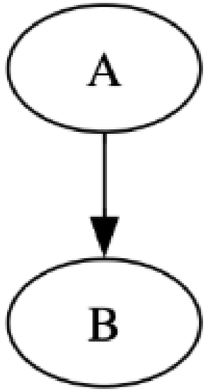
ABA

~~c.~~



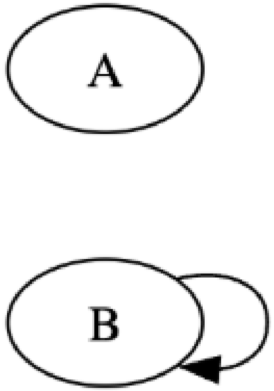
ABA

d. ✓



?

~~e.~~



BB

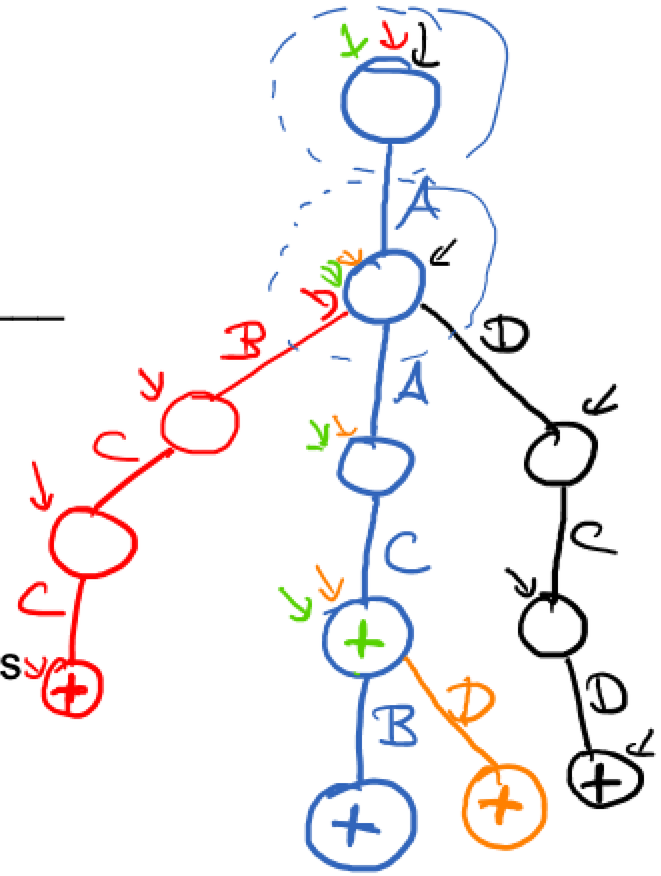
The following strings are inserted into an empty trie in the given order:
AACB, AACD, AAC, ABCC, ADCD
Trace through the insertions, then answer the following questions about the trace.

A. How many nodes does the trie contain after all 5 strings are inserted? 12

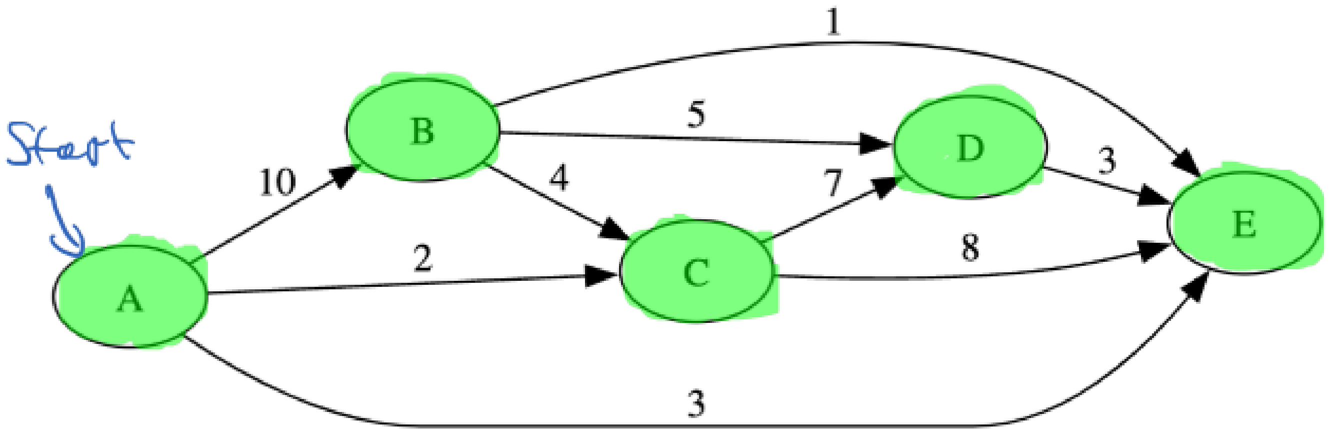
B. How many children does the root node of the trie have after all 5 strings are inserted? 1

C. What is the highest number of children of any node in the trie after all 5 strings are inserted? 3

D. Which of the strings can be inserted without changing the number of nodes in the trie? AAC



Trace through Dijkstra's algorithm with starting node A on the following graph. While tracing through the algorithm, list the order in which it visits the nodes of the graph.



node	pred	cost
A	null	0
B	A	10
C	A	2
D	C	9
E	A	3

Visiting order of nodes: A, C, E, D, B

p.q.: ~~[A, null, 0]~~, ~~[B, A, 10]~~, ~~[C, A, 2]~~, ~~[E, A, 3]~~, ~~[D, C, 9]~~, ~~[E, C, 10]~~

Fill in the blank in the following HTML element so that the style rule applies to it.

```
.course { color: green; font-weight: bold; }
<li class="course">CS400</li>
```

A. name="course"
B. class="course"
C. id="course"
D. type="course"
E. value="course"

We implement Counting Sort for single digit numbers between 0-8 (inclusive), and then use it to sort the input sequence: 8, 3, 3, 1, 5. What's the size of the longest array that we'll need to instantiate during the execution of Counting Sort on this data?

input array: 5

counts array: 9

endpos array: 9

output array: 5

Which of the strings entirely matches the following regular expression?

`[ab]+x?c*`

xcc

aaaaaaxc

bbxbbx

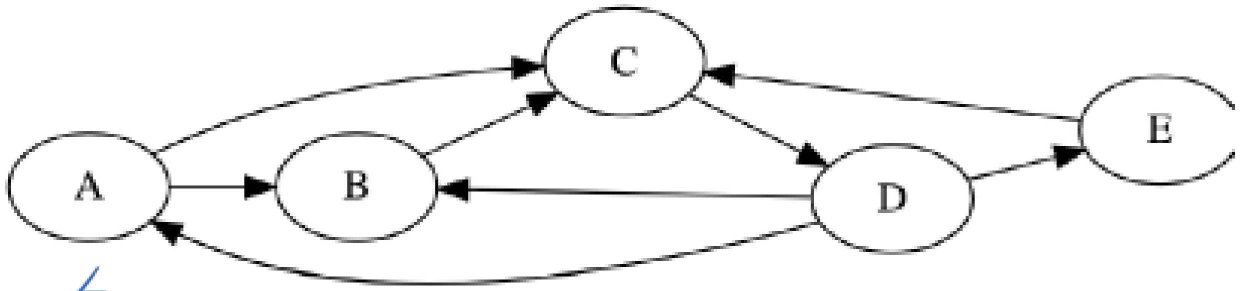
axxc

x

Regex Reference:

```
[ab] either a or b
+    one or more repetitions
?    zero or one repetitions
*    zero or more repetitions
```

Which of the following descriptors describe the graph below? Select all that apply.



A. Directed
B. Weighted
C. Weakly Connected
D. Strongly Connected
E. Cyclic

Suppose that the following code is run, and then an http request is sent to this webserver. What order would you expect the print statements within this webserver to print out when this happens?

```
HttpServer server = new HttpServer(new InetSocketAddress(8080),0);
System.out.print("X");
server.createContext("/").setHandler( x -> System.out.print("Y") );
System.out.print("Z");
server.start();
```

A. XYZ
B. XZY
C. YXZ
D. YZX
E. ZYX

Add both a button with the text “reveal answer” on it and a paragraph starting with “the answer is: ” to the following HTML document. When the user clicks the button, the browser should run the function “revealAnswer”, which adds the answer it receives from code running on the webserver to be visible near the bottom of the page.

```
<!DOCTYPE html>
<html>
    <head>
        <title>Answer Revealer</title>
        <script>
            let revealAnswer = () => {
```

equivalent to: function revealAnswer() {

```
                fetch("http://12.10.144.122/getAnswer")
                    .then( (answer) => answer.text() )
                    .then( (answer) => {
                        document.querySelectorAll("#output")[0]
                                                .innerHTML = answer.value;
                    } );

            }
        </script>
    </head>
    <body>
<p>the answer is: <span id="output"></span></p>
<input type="button"
       value="reveal answer"
       onclick="revealAnswer()" />

            </body>
        </html>
```

. What numbers in which order does the file `output.txt` contain after this Bash command is run in a folder that contains the following file `list.txt`?

```
grep -P "^1" list.txt | sort -r > output.txt
```

16
15

16
15

list.txt:

```
16
15
33
2
```

a. 16 15
b. 15 16
c. 16 15 33 2
d. 16 2 15 33
e. 33 15 2 16

Regular Expressions Reference:

`0-9`	`matches the character`
`^`	`matches the beginning of a line or string`

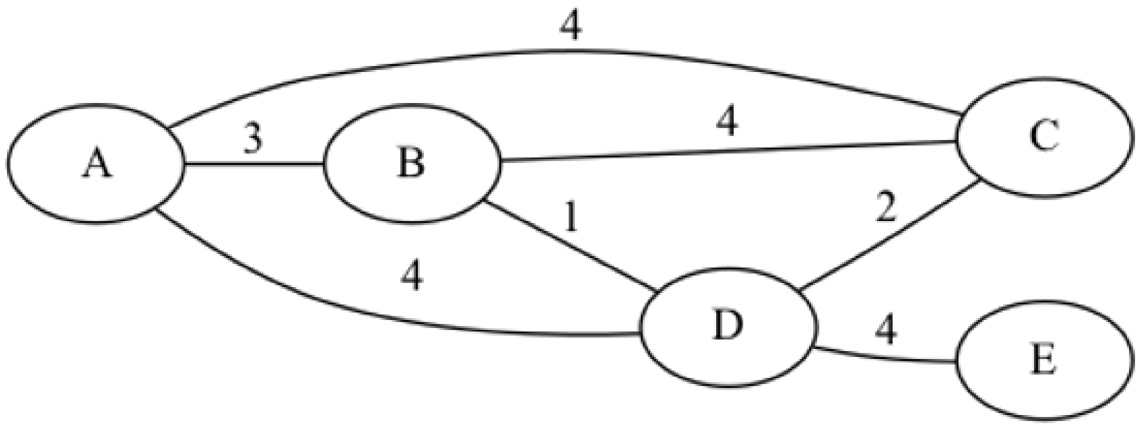
The following hash table of capacity 10 uses open addressing with linear probing and the following hash function: hash(x) = x % capacity.

index:	0	1	2	3	4	5	6	7	8	9
key:	10	20	51	109						99

Which of the keys in the hash table was inserted last?

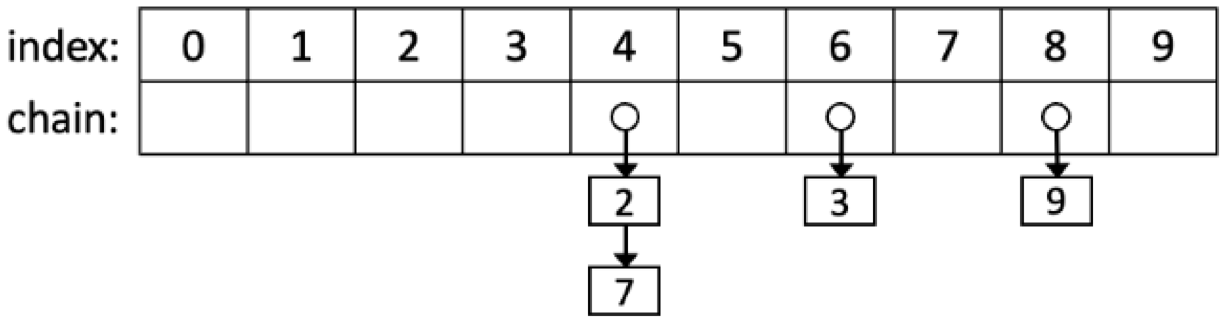
a. 99
b. 10
c. 20
d. 51
e. 109

When applying Prim's algorithm to the following graph, it generates a set of four edges, adding edges with the following weights and in the following order to the set: 3, 1, 2, 4. What was the starting node for the algorithm?



a. A
b. B
c. C
d. D
e. E

The following hash table of capacity 10 uses chaining with linked lists.



Which of the hash functions could have been used to create the above table?

a. hash(x) = x % capacity
b. hash(x) = (x * 2) % capacity
c. hash(x) = floor(x / 2) % capacity
d. hash(x) = (x + 1) % capacity
e. hash(x) = (x * x) % capacity